Team Name: _____ Team #: ____ Score: 200 / 200

Competitor Name(s): _____

Section A 10/10 points

1. Nitrous oxide (1) (1)
2. Infrared (1) (1)
3. Absorption (1) spectrum (1) (2)
4. It is strongly (1) absorbed (1) by N_2O (1) (3)
5. Water vapor (2) (2)
6. Vibration (1) (1)

Section B 10/10 points

7. 1.0001 (1) (1)
8. 29.99 to 30.01 km/s (4) (4)
9. 0.46 to 0.47 km/s (3) (3)
10. 29.52 to 29.55 km/s (2) (2)

Section C 20/20 points

11. 430 nm and 660 nm (1 each, I'll accept within 10) (2) (2)
12. For the blue band, water with lowest chlorophyll content has a radiance 1000 times higher than water with highest chlorophyll content. (1) That goes down to 10 times higher in the green band, meaning there is significantly more absorption of blue light in chlorophyll-rich samples. (2) (3)
13. Phytoplankton/productivity (1) (1)
14. You see the same place on Earth at the same time everyday, provide good global coverage, allow for high latitude measurements. (1)
15. Continental rise/slope pushes cold water currents to the surface from the deepest portions of the ocean, carrying increased phytoplankton concentrations to the surface. OR: nutrient runoff/pollution from the nearby land (2)
16. Heightened concentrations (1), sun synchronous orbits measure a narrow swath of the earth and cannot measure it all at once due to low their orbit radius (3)
17. The depth at which 90% of upwelling irradiance is originated above OR where downwelling irradiance attenuates to 10% of surface values. (2)
18. SCMLs represent a decoupling of the surface and subsurface concentrations of Chl-a, so not accounting for them would produce a significant underestimation of Chl-a content in a vertical column. (3)
19. Surface reflectance and temperature can be used as constraints along with light attenuation increasing with depth to model its shape and topography. (2) (3)

Section D 20/20 points

20. Shortwave (1) (1)
21. Sun (2) (2)
22. Albedo (2) (2)
23. High level (1) of cloud cover (1) (2)
24. Air rises here (1), leaving behind moisture (1) (2)
25. Longwave (1) (1)
26. High (1) (1)
27. Mountains (1), Rainforest/Jungle (1), Polar (1) (3)
28. Mountains because of low temperatures (2), Rainforest/jungle because of frequent (1) cloud cover (1), polar areas because of cold/high altitude (1) cloud cover (1) (6)

Section E 25/25 points

29. Increase (1)
30. It is hindered by scattering/greater uncertainty in CO2 measurements. (2)
31. MODIS (1)
32. Clear sky XCO2/baseline carbon dioxide concentration (2)
33. The center of the forest fire/where there is the most smoke (either acceptable, 1pt for smoke) (2)
34. Reflectance is reduced due to loss of vegetation cover, marking an area having experienced a fire. (2)
35. Increased surface temperature. (2 points) Measured by 3.5-4μm bandwidth (1 point) (3)
36. Yields significant increases in SWin (2)
37. Latent heat flux would be increased as a result of decreased aerosol concentrations (2)
38. Increased latent heat would lead to increased plant/canopy transpiration, which leads to dryer soil. This, in turns, leads to greater susceptibility to forest fires. (3)
39. Melting of soil ice may increase soil moisture (2)
40. Aerosol concentration has a greater impact (2 points), forest fires made the most likely by increased aerosol concentration/ mitigation (1 point). (3)

Section F 25/25 points

41. Passive (1) (1)
42. Degrees Celsius (2) (2)
43. No (1) (1)
44. Land (2) (2)
45. No observations (2) because of the combination of swath width (1) and orbital period (1) (4)
46. Clouds (2) prevent observations of the sea at this wavelength (1) (3)
47. 2200 to 2500 km (3) (3)
48. Polar/Sun synchronous (1) (1)
49. 93 to 102 minutes (2) (2)
50. SWIR is at shorter wavelengths than the peak (1), while LWIR is at longer wavelengths (1) (1 pt for opposite sides) (2)
51. The shorter limb/shortwave side of an emission spectrum varies more with temperature (2) which means the SWIR band is more sensitive (2) (4)

Section G 25/25 points

52. Bjerknes feedback (2)
53. It increases sea surface temperature in the central Pacific (also acceptable: creates more sea surface temperature anomalies in the central Pacific) (2)
54. Atmospheric convection is shifted to the Central Pacific (1 point), weakening trade winds (1 point), and reinforcing the initial response triggered by stochastic wind variation/causing a positive feedback loop (1 point) (3)
55. Warmer climates (1 point); the feedback strength is directly connected to development of SST anomalies, so higher values at 0 ka compared to 21 ka suggest warmer climates connect to more sea surface temperature anomalies (3)
56. Inversely related (2)
57. Directly related/higher feedback, more extreme events (2)
58. Sufficient wind variation and fluctuation is required to initiate the feedback loop (2 points), these can't happen in glacial periods but are easily activated in elevated greenhouse gas periods (2 pts) (4)
59. It has moved west (1)
60. Background sea surface temperatures increased in the western tropical Pacific (2 points), promoting enhanced upward motion in the overlying atmosphere. This leads to increased precipitation there (2 points) (4)
61. More frequent, shorter period cycles (1 pt each) (2)

Section H 30/30 points

62. Decreases (1)
63. 0.05 m (3)
64. Counting people not in crowds, distinguishing the fronts from rears of cars. Features of similar size accepted (3)
65. Higher temporal resolution would allow for mitigation of the data gaps resulting from cloud cover or glint (1 pt). Also may allow for better study of phenomena which happen on a much shorter time scale (such as tides) (2 for description) (3)
66. Change from a polar orbit to a geostationary orbit (1 pt for its orbit) (2)
67. Rainfall results more from higher altitude clouds, which are colder and thus lower in temperature. Therefore, a colder cloud top can be considered a likely place of rainfall in the near future. (3)
68. Passive (1)
69. Microwave has longer wavelength (1 point) which allows for imaging inside the cloud, as opposed to just the cloudtop (2 points) (3)
70. Significantly higher spatial resolution (2)
71. IR can be placed at significantly higher elevations, and thus can be used in a geostationary orbit. (2) Microwave needs to be significantly closer and thus is typically used in polar orbiting satellites. (2) (4)
72. 81.3 degrees (2)
73. 101 minutes (2)
74. Microwave (1)

Section I 35/35 points

75. Stefan-Boltzman law (1) (1)
76. 63.5 to 64.5 MW/m^2 (3) (3)
77. 45,800 to 46,000 (3) (3)
78. 1370 to 1420 W/m^2 (2) (2)
79. 4 (3) (3)
80. 278 to 282 K (4) (4)
81. The Earth reflects (1) 30% of shortwave/Solar radiation (1) (2)
82. Decrease (1) (1)
83. 255 to 258 K (5) (5)
84. $Q_A = Q_E/2$ (2) (2)
85. $Q_E = Q_A + 0.7(Q_S/4)$ (3) (3)
86. 303 to 306 K (6) (6)

MIT Invitational - Repaired practice edition - Answer key

Division C | Remote Sensing | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

8. (4 points) Authored replacement for source item Question 8: Why start with a question rather than an index?

Unit: Remote sensing foundations and history.
Worked-case data: An image becomes brighter after a display stretch, but its stored reflectance values are unchanged. Has surface reflectance increased?

answer: The question determines the needed observable, wavelengths, resolution, dates, and validation; a convenient ratio may not discriminate the alternatives. (4 points).

10. (2 points) Authored replacement for source item Question 10: What does a sensor directly record in optical imaging?

Unit: Remote sensing foundations and history.
Worked-case data: An image becomes brighter after a display stretch, but its stored reflectance values are unchanged. Has surface reflectance increased?

A. Tree species with certainty

B. Human land-use intent

C. A complete climate mechanism

D. A band-dependent energy response

Correct: D

16. (2 points) Authored replacement for source item Question 16: Which sensor is active?

Unit: Sensors, radar, lidar, and illumination.
Worked-case data: A lidar pulse has round-trip time 8 microseconds. Use c = 3 x 10^8 m/s.

A. A natural-light photograph

B. Radar

C. A reflected-sunlight camera

D. A passive thermal radiometer

Correct: B